5.8 Coordinate Proofs

Name _____ Period ____ Date _____

Geometry 5.8 (30-minute plan) · Student sheet

I can…

- place a figure in the coordinate plane in a convenient way.
- use the Distance and Midpoint Formulas and slopes in a coordinate proof.
- write a coordinate proof with variable coordinates.

Vocabulary · fill in the missing word

Coordinate proof — a proof that places a figure in the coordinate plane and uses coordinates and _____

Part 1 · Notes and examples (fill in each note, then work the example)

Placing figures: put a vertex at the _____ and sides along the axes when you can.

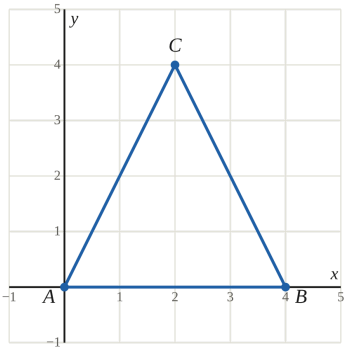
Example 1

Place a 6-by-4 rectangle in the coordinate plane in a convenient way. Give the vertices.

Tools: the Distance Formula shows segments are congruent, the Midpoint Formula finds midpoints, and _____ show segments are parallel or perpendicular.

Example 2

Prove that △ABC with A(0, 0), B(4, 0), and C(2, 4) is isosceles.



Variable coordinates: use letters like (2a, 0) and (0, 2b) to prove a statement for _____ figures of that type.

Example 3

Right △OAB has O(0, 0), A(2a, 0), and B(0, 2b). M is the midpoint of $\overline{AB}$. Prove that M is the same distance from O, A, and B.

Part 2 · Partner practice

Partner practice

Is the quadrilateral with vertices (0, 0), (3, 0), (4, 2), and (1, 2) a parallelogram? Use slopes.